Have You Ever Noticed?

- In the past few years, I have noticed that summers have become hotter and winters warmer in Elko. Do you feel the same?
- This made me wonder: Is Nevada's climate truly getting warmer? Is Elko experiencing a larger temperature change? What about Reno and Las Vegas?
- To answer these questions, I decided to analyze historical temperature data.

Background

- **Global Warming -** refers to the long-term increase in Earth's average temperature due to greenhouse gases.
- **Urban Heat Island Effect -** cities are often warmer than rural areas because buildings, roads, and reduced vegetation absorb and retain heat.
- **Climate Normals -** climate scientists usually use 30-year averages called climate normals to understand long-term climate trands.
- **Heating and Cooling Degree Days (HDD/CDD) -** measure how much heating or cooling is needed based on temperature. They are used to estimate energy demand.

Objectives

- **This project has three objectives:**

1. To explore changes in **annual average temperatures** over the past 50 years.
2. To compare **the rates of temperature, increase** in Elko, Reno, and Las Vegas.
3. To investigate how temperature changes affect **energy demand** in daily life in Elko.

Research Questions (RQs)

- **RQ1:**

  Are the annual average temperatures in Elko, Reno, and Las Vegas increasing over the past 50 years?

- **RQ2:**

  Which city has the highest rate of temperature increase: Elko, Reno, or Las Vegas? *(Sprint vs. Marathon)*

- **RQ3:**

  How do temperature changes affect heating and cooling demand in Elko?

Hypothesis

- **Hypothesis 1:**

  The annual average temperatures in Elko, Reno, and Las Vegas will show an increase over the past 50 years.

- **Hypothesis 2:**

  Las Vegas has the highest rate of temperature increase due to urbanization and the urban heat island effect.

- **Hypothesis 3:**

  Heating demand in Elko will decrease over time, while cooling demand will increase.

Data Source

- Daily maximum and minimum temperature data (1975–2025) were downloaded from the **National Centers for Environmental Information (NOAA).** https://www.ncei.noaa.gov/cdo-web/datatools
- Three separate **CSV data files** were obtained for **Elko, Reno, and Las Vegas.**
- Each file includes **Date,** Daily Maximum Temperature **(T_{max}),** and Daily Minimum Temperature **(T_{min}).**
- All temperature values were recorded in degrees **Fahrenheit (°F)**.

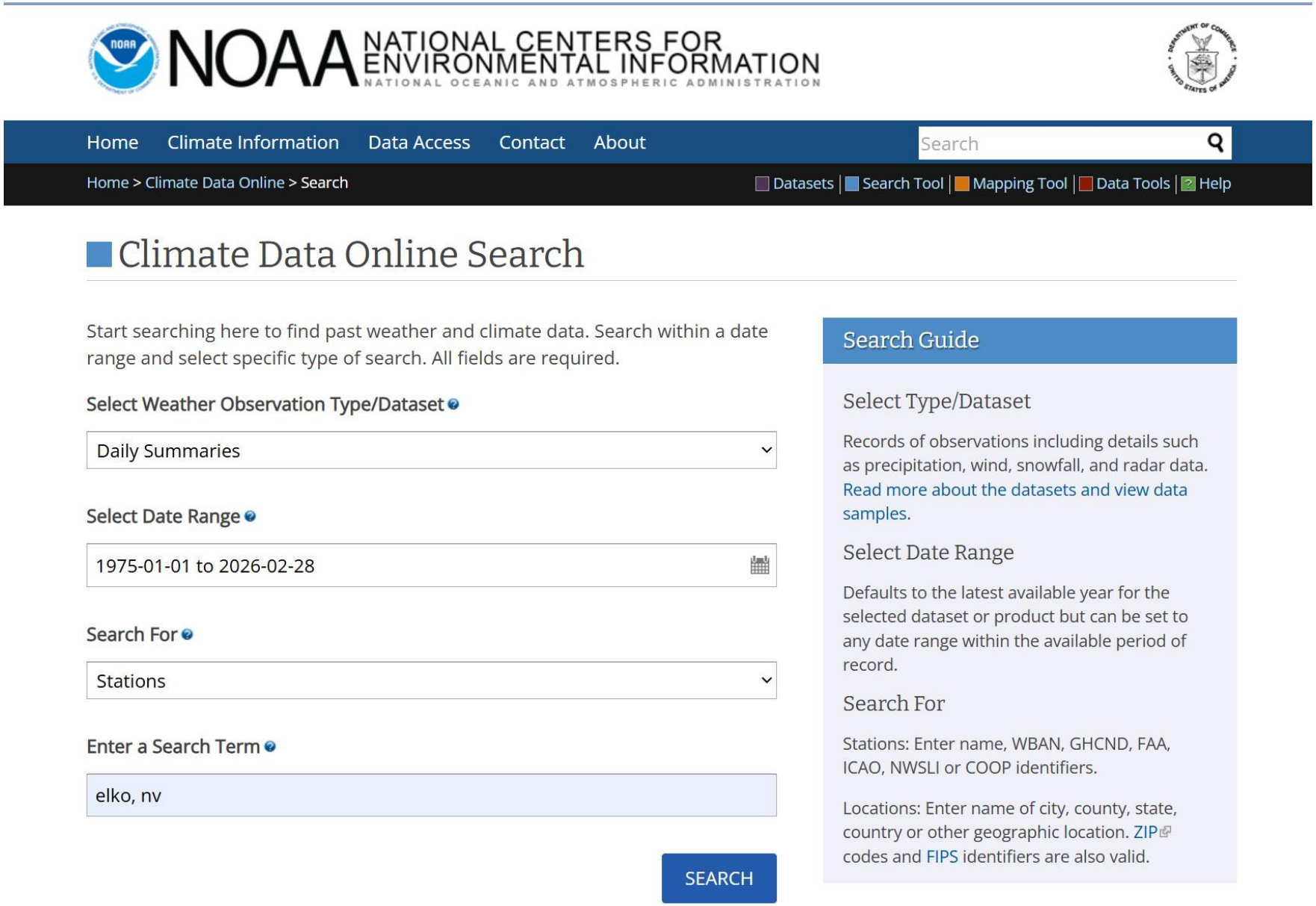


Figure: NOAA Climate Data Online tool used to download daily temperature data.

Procedure

- **Step 1:** Download daily maximum and minimum temperature data (1975-2025) from NOAA for **Elko, Reno, Las Vegas**.
- **Step 2:** Calculate **annual average temperatures** and **warming rates**, and generate **line graphs**, using Microsoft Excel.
- **Step 3:** Compare warming rates for two time periods: 1975–2025 (**Marathon**) and 2000–2025 (**Sprint**) in Desmos.
- **Step 4:** For Elko, calculate daily Heating Degree Days (**HDD**) and Cooling Degree Days (**CDD**) using a base temperature of 65°F, then sum HDD and CDD for each year.
- **Step 5:** Analyze the results and discuss how they **answer the research questions**.

Data Processing

- Daily Average Temperature:

$$T_{day_avg} = \frac{T_{max} + T_{min}}{2}$$

- Monthly Average Temperature:

$$T_{month_avg} = \frac{\sum T_{day_avg}}{n}$$

(*Note:* $\sum T_{day_avg}$ *is the sum of* T_{day_avy} *and n is the number of days.*)

- Annual Average Temperature:

$$T_{year_avg} = \frac{\sum T_{month_avg}}{12}$$

(*Note:* $\sum T_{month_avg}$ *is the sum of* T_{month_avg})

- Rate of Temperature Change (Slope):

$$Rate = \frac{Change\ of\ Temperature}{Change\ of\ year}$$

- Heating & Cooling Degree Days (Base 65°F)

$$HDD = \max\ (0, 65 - T_{day_avg})$$

$$CDD = \max\ (0, T_{day_avg} - 65)$$

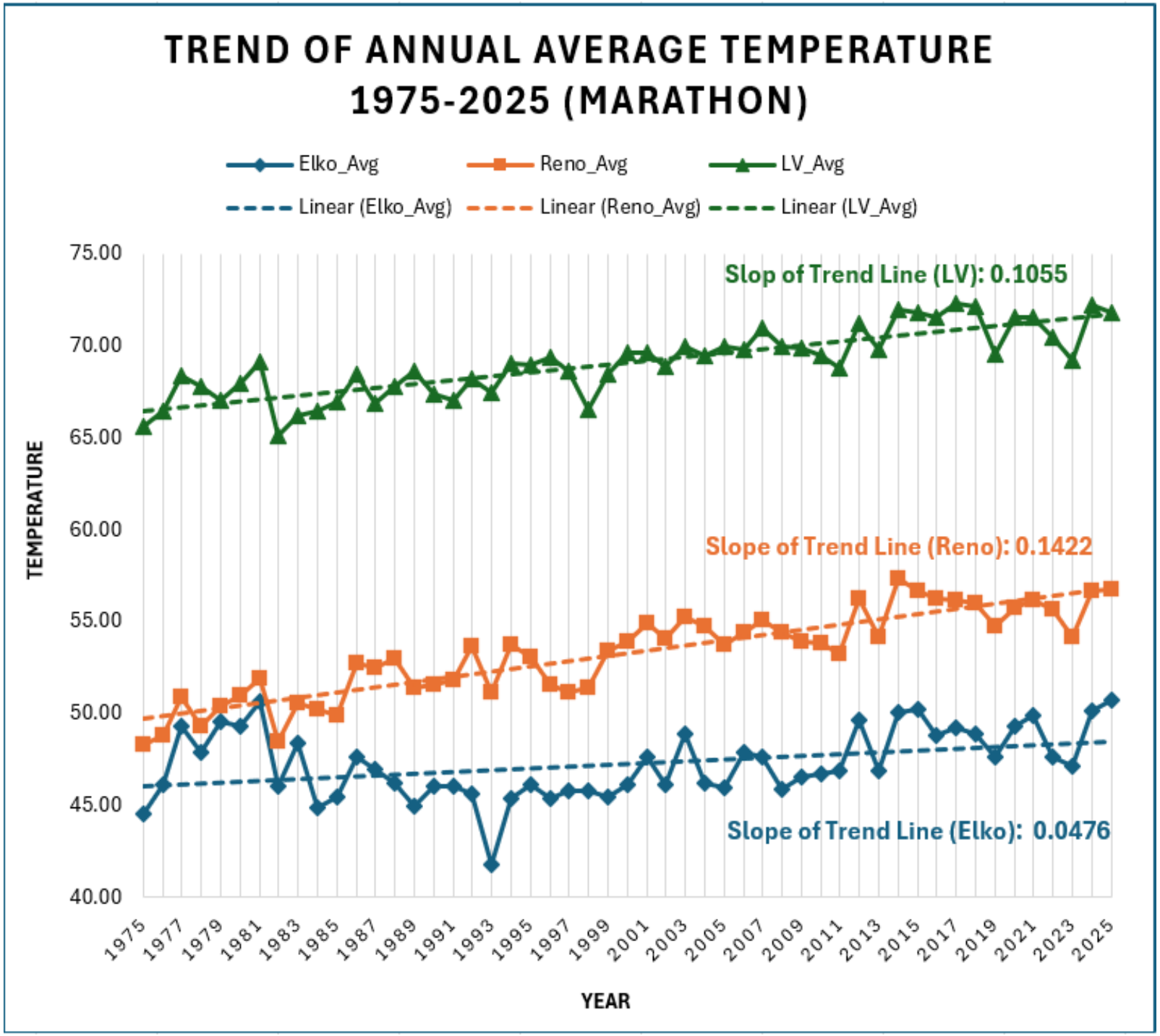
(*Note: the MAX function means "choose the largest value."*)

Data Analysis and Results

Marathon (1975-2025)

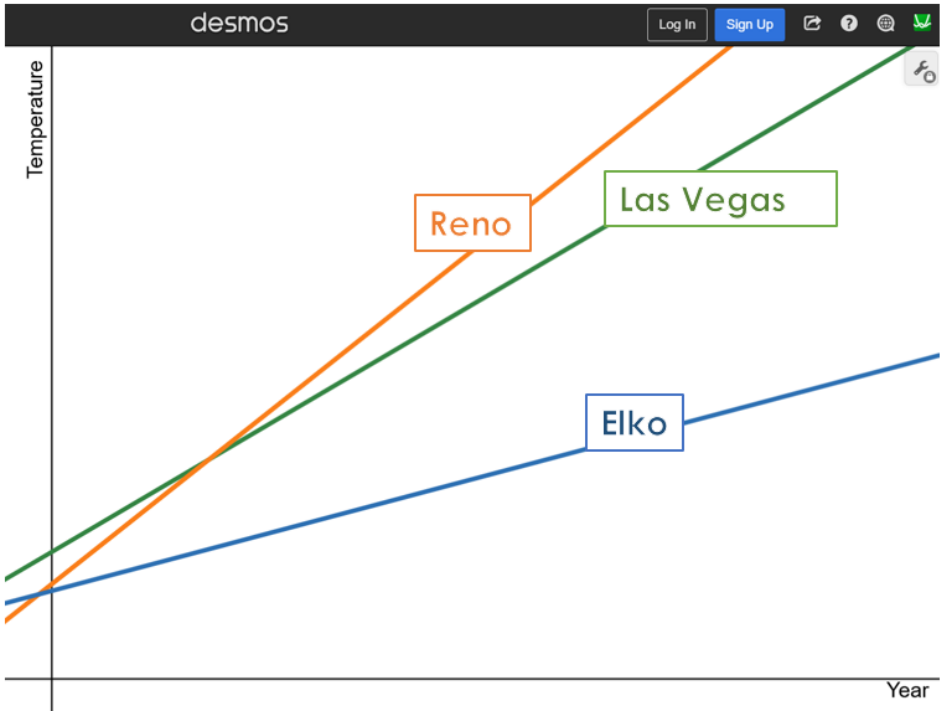
Sprint (2000-2025)

Changes of Energy Demand

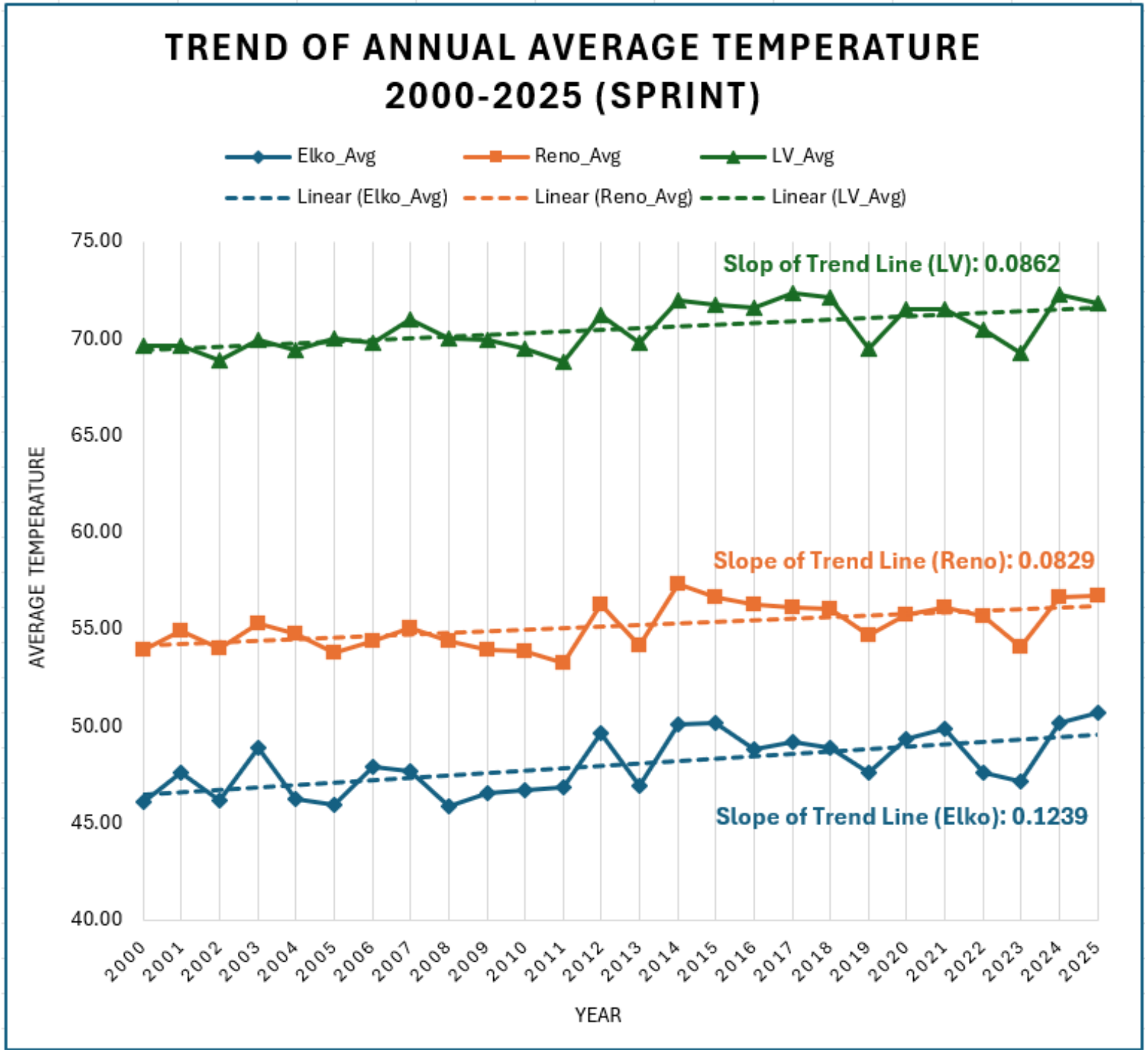

TREND OF ANNUAL AVERAGE TEMPERATURE
1975-2025 (MARATHON)
Elko_Avg
Reno_Avg
LV_Avg
Linear (Elko_Avg)
Linear (Reno_Avg)
Linear (LV_Avg)
Slop of Trend Line (LV): 0.1055
Slope of Trend Line (Reno): 0.1422
Slope of Trend Line (Elko): 0.0476
TEMPERATURE
75.00
70.00
65.00
60.00
55.00
50.00
45.00
40.00
1975
1977
1979
1981
1983
1985
1987
1989
1991
1993
1995
1997
1999
2001
2003
2005
2007
2009
2011
2013
2015
2017
2019
2021
2023
2025
YEAR

	Annual Average Temperature (1975)	Average Average Temperature (2025)	Approx. Range (Min–Max)	Rate of Increase (Slope of Trendline) (°F per year)	Warming Speed (Rank)
Elko	44.5 °F	50.5 °F	~42 – 51 °F	0.0476	3
Reno	48.5 °F	56.5 °F	~48 – 57 °F	0.1422	1
Las Vegas	65.5 °F	72.0 °F	~65 – 73 °F	0.1055	2

- The line chart shows that the annual average temperatures in Elko, Reno, and Las Vegas have steadily increased from 1975 to 2025.
- In the marathon of Nevada's warming race, Reno is the winner.

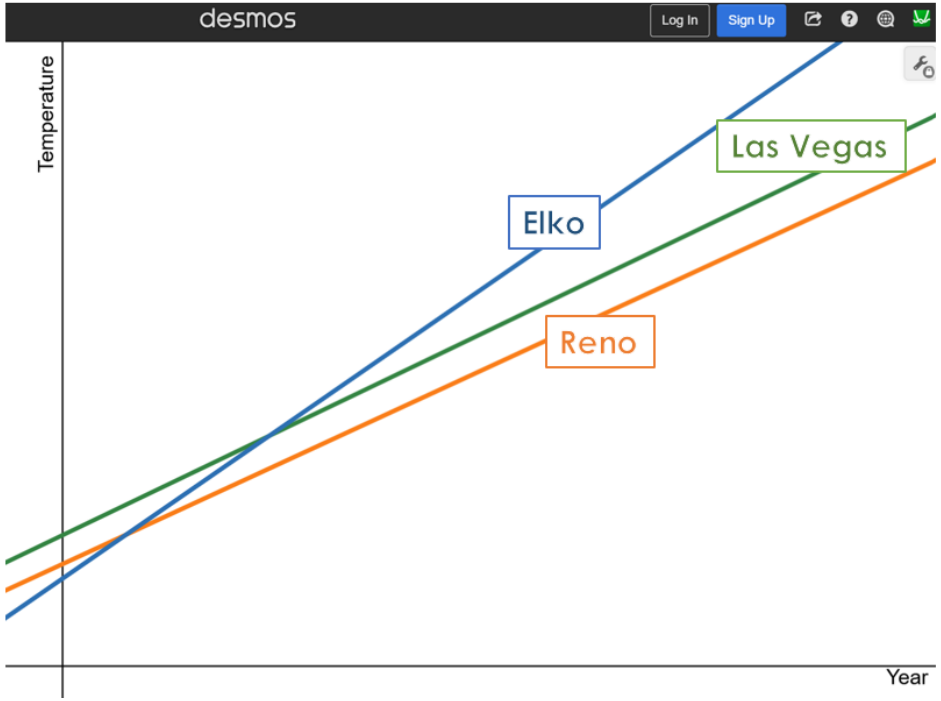


The trendlines for three cities were graphed using Desmos.


TREND OF ANNUAL AVERAGE TEMPERATURE
2000-2025 (SPRINT)
Elko_Avg
Reno_Avg
LV_Avg
Linear (Elko_Avg)
Linear (Reno_Avg)
Linear (LV_Avg)
Slop of Trend Line (LV): 0.0862
Slope of Trend Line (Reno): 0.0829
Slope of Trend Line (Elko): 0.1239
AVERAGE TEMPERATURE
75.00
70.00
65.00
60.00
55.00
50.00
45.00
40.00
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
YEAR

	Annual Average Temperature (2000)	Average Average Temperature (2025)	Approx. Range (Min–Max)	Rate of Increase (Slope of Trendline) (°F per year)	Warming Speed (Rank)
Elko	46.1 °F	50.7 °F	~45.9 – 50.7 °F	0.1239	1
Reno	53.9 °F	56.7 °F	~53.3 – 57.3 °F	0.0829	3
Las Vegas	69.6 °F	71.8 °F	~68.8 – 72.3 °F	0.0862	2

- In the sprint of Nevada's warming race, Elko is the fastest.



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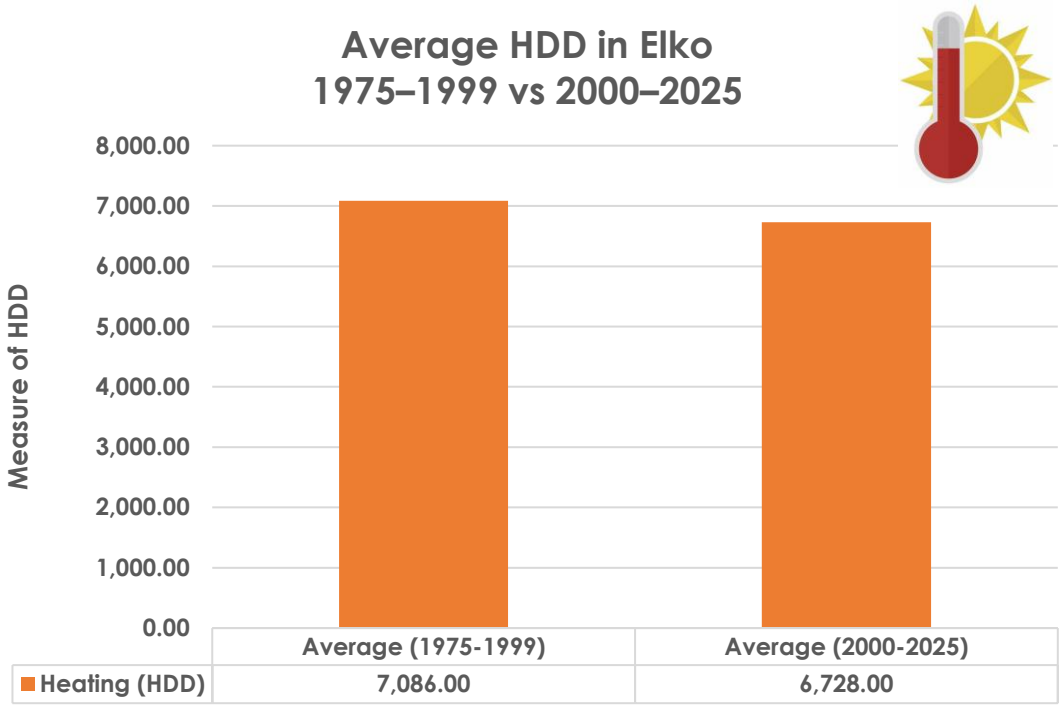
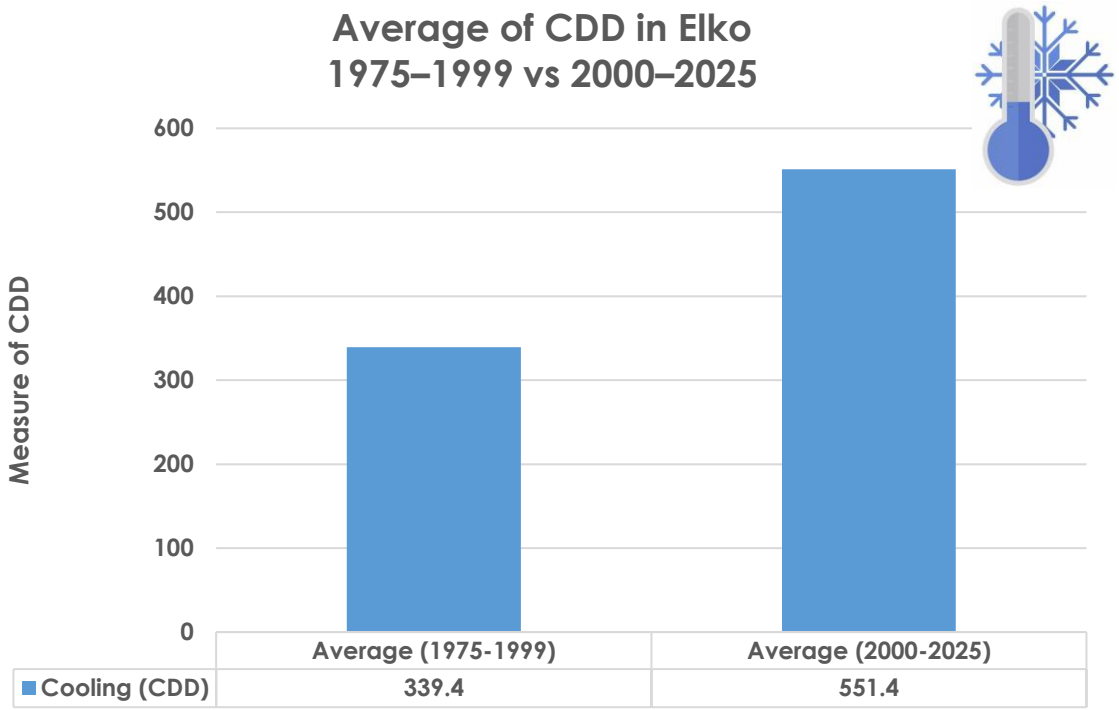
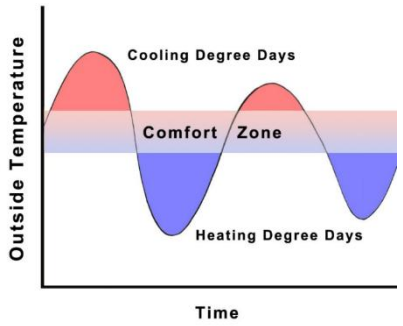

Average HDD in Elko
1975–1999 vs 2000–2025
Measure of HDD
8,000.00
7,000.00
6,000.00
5,000.00
4,000.00
3,000.00
2,000.00
1,000.00
0.00
Average (1975-1999)
Average (2000-2025)
Heating (HDD)
7,086.00
6,728.00


Average of CDD in Elko
1975–1999 vs 2000–2025
Measure of CDD
600
500
400
300
200
100
0
Average (1975-1999)
Average (2000-2025)
Cooling (CDD)
339.4
551.4

Changes in Heating and Cooling Demand in Elko

Metric	1975–1999 Average	2000–2025 Average	Change	Percent Change	Interpretation
Heating Degree Days (HDD)	~7,086	~6,728	−358	−5.1%	Less heating needed
Cooling Degree Days (CDD)	~339.4	~551.4	212	0.625	More cooling needed

- Heating demand in Elko has slightly decreased over time, while cooling demand has increased significantly, showing how warming temperatures affect energy use in daily life.



Conclusion

- RQ1: Annual average temperatures in Elko, Reno, and Las Vegas increased from 1975 to 2025, showing a clear warming trend in Nevada.
- RQ2: Reno warmed the fastest over the 50-year period, while Elko warmed the fastest in the recent 25 years.
- RQ3: In Elko, Heating Degree Days (HDD) decreased and Cooling Degree Days (CDD) increased, meaning less heating and more cooling are needed over time.

Takeaways & Questions

- All three cities warmed over the past 50 years, showing a clear long-term warming trend in Nevada.
- Reno is the Marathon race winner, but Elko wins the Sprint race.
- Do you think energy bills at home will increase if temperatures continue to rise?

Data is here!